HU Collection

Engineering and Science Admissions Test

Mock Set F

Mathematics 1

Time allowed: 40 minutes
Questions: 27 multiple choice
Calculators: not permitted
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Name: Date sat: Score: / 27

Instructions

Answer all 27 questions. Each question has exactly one correct option and is worth one mark. No marks are deducted for wrong answers, so answer every question. Calculators are not permitted. This paper is deliberately longer than the time allows: move on from any question that is costing you more than two minutes, and come back to it at the end.

Q	1	2	3	4	5	6	7	8	9	10
Ans										

Q	11	12	13	14	15	16	17	18	19	20
Ans										

Q	21	22	23	24	25	26	27			
Ans										

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1. A conservator is covering the outer curved surface of a hemispherical dome of radius 3 m with gold leaf. The leaf has a uniform thickness of $0.125\,\mu$m, where $1\,\mu\text{m} = 10^{-6}$ m, and the density of gold is 19.2 g/cm^3.

Ignoring overlaps and waste, what mass of gold is needed?

A 43.2π g **B** 86.4π g **C** 4.32π g

D 432π g **E** 64.8π g **F** 21.6π g

2. A microbiologist estimates the number of bacteria in a culture by serial dilution. She transfers 1 ml of the culture into 99 ml of sterile water and mixes thoroughly. She then transfers 1 ml of this mixture into another 99 ml of sterile water and mixes, and finally she transfers 0.5 ml of the second mixture into 4.5 ml of sterile water and mixes.

She spreads 0.05 ml of the final mixture on a dish, and 63 colonies grow, each from a single bacterium. On this evidence, approximately how many bacteria were there in each litre of the original culture?

A 1.26×10^{12} **B** 1.26×10^{8} **C** 3.15×10^{8}

D 1.26×10^{11} **E** 6.3×10^{10} **F** 1.26×10^{10}

3. Kleiber's law states that the daily energy requirement of a mammal is proportional to $M^{3/4}$, where M is its body mass. Assume that the law holds, with the same constant of proportionality, for two species: one whose animals each have mass 81 kg, and one whose animals each have mass 16 kg.

A reserve can hold either a herd of the larger animals or a herd of the smaller animals, the two herds having the same total mass. What is the ratio of the total daily energy requirement of the herd of smaller animals to that of the herd of larger animals?

A 1 : 1 **B** 3 : 2 **C** 9 : 4

D 2 : 3 **E** 27 : 8 **F** 81 : 16

4. At noon, ship P is 10 km due north of ship Q. Ship P sails due east at a constant 3 km/h, and at the same moment ship Q sets off due north at a constant 4 km/h. Treat the sea as a flat plane.

At what time are the ships closest together, and how far apart are they then?

A at 14:30, 7.5 km apart **B** at 13:00, $3\sqrt{5}$ km apart **C** at 13:36, 8 km apart

D at 13:36, 6 km apart **E** at 14:00, $2\sqrt{10}$ km apart **F** at 13:20, 6 km apart

5. Two small loudspeakers A and B are 6 m apart. The sound intensity that a loudspeaker produces at a point is directly proportional to its power and inversely proportional to the square of the distance of the point from the loudspeaker. Loudspeaker B has 4 times the power of loudspeaker A.

Consider the points on the straight line through A and B, extended indefinitely in both directions, at which the intensities from the two loudspeakers are equal. What is the greatest distance between two such points?

A 2 m **B** 8 m **C** 10 m

D 4 m **E** 6 m **F** 12 m

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6. The table shows the number of goals scored by the home team in each of 200 football matches.

Goals	0	1	2	3	4
Matches	30	50	60	40	20

The results of k further matches are then added to the data, and in each of these matches the home team scored 0 goals. What is the smallest value of k for which the median number of home goals, over all $200 + k$ matches, is 1?

A 41 **B** 40 **C** 20

D 42 **E** 60 **F** 21

7. A carpenter has a plank whose length is 3.2 m, correct to the nearest 0.1 m. She cuts shelves from it, all of the same length, which is 40 cm correct to the nearest centimetre, and she cuts as many shelves as she can. Ignore the width of the saw cuts.

Let G be the greatest number of shelves that she can be certain of obtaining, and let H be the greatest number of shelves that she could possibly obtain. What are G and H?

A $G = 7$ and $H = 7$ **B** $G = 8$ and $H = 9$ **C** $G = 6$ and $H = 8$

D $G = 8$ and $H = 8$ **E** $G = 7$ and $H = 8$ **F** $G = 7$ and $H = 9$

8. The twelve hour marks on a clock face are equally spaced around a circle. Three straight chords are drawn: one joining the marks for 12 and 4, one joining the marks for 1 and 5, and one joining the marks for 2 and 11. The three chords form a small triangle inside the circle.

What is the size of the largest angle of this triangle?

A 105° **B** 75° **C** 135°

D 120° **E** 90° **F** 150°

9. A square array of 36 dots is arranged in 6 rows and 6 columns, with neighbouring dots 1 cm apart horizontally and vertically. Two different dots are chosen at random, with every pair of dots equally likely.

What is the probability that the distance between the two chosen dots is a whole number of centimetres?

A $\frac{1}{7}$ **B** $\frac{32}{105}$ **C** $\frac{34}{105}$

D $\frac{38}{105}$ **E** $\frac{2}{7}$ **F** $\frac{1}{3}$

10. A data set has mean 45, median 47, range 40, lower quartile 38 and upper quartile 54, where the quartiles are the medians of the lower and upper halves of the ordered data. Every value x in the data set is replaced by the coded value $ax + b$, where a and b are constants with $a \neq 0$, chosen so that the coded values have mean 60 and range 50.

Which of the following statements are true?

I. The median of the coded values could be 57.5.

II. The interquartile range of the coded values must be 20.

III. The lower quartile of the coded values must be 51.25.

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A I and II only **B** II and III only **C** II only

D I only **E** I, II and III **F** I and III only

11. A patient receives a dose of a drug every 2 hours, starting at time 0, when the concentration of the drug in the blood is zero. Each dose instantly increases the concentration by 18 mg/l. Between doses the concentration decreases in such a way that at the end of each hour it is 80% of what it was at the start of that hour.

After many doses, the concentration just before each dose settles down to a fixed value. What is this value, in mg/l?

A 90 **B** 45 **C** 32

D 27 **E** 50 **F** 72

12. A data logger records the temperature, in °C, at the end of each of ten consecutive hours, giving readings $T_1, T_2, \ldots, T_{10}$. Because of a fault, it outputs only the seven 4-point moving averages

$$\frac{T_1+T_2+T_3+T_4}{4},\quad \frac{T_2+T_3+T_4+T_5}{4},\quad \ldots,\quad \frac{T_7+T_8+T_9+T_{10}}{4},$$

which are, in order,

$$12,\quad 13,\quad 15,\quad 14,\quad 16,\quad 18,\quad 17.$$

A technician also knows that $T_1 = 9$, $T_2 = 11$ and $T_3 = 14$.

What is the median of the ten readings?

A 13 **B** 14.5 **C** 16

D 14 **E** 15 **F** 13.5

13. Three identical steel balls, each of radius 3 cm, rest on a horizontal table, each touching the other two. A smaller ball rests on the table in the gap between them and touches all three of the larger balls.

What is the radius of the smaller ball, in cm?

A $\sqrt{3}-1$ **B** 1 **C** $\frac{3}{4}$

D $2\sqrt{3}-3$ **E** $3-\sqrt{3}$ **F** $\frac{3}{2}$

14. In triangle OAB, the point P lies on the side AB with $AP : PB = 2 : 1$, and Q is the midpoint of OP. The straight line through A and Q meets the side OB at R.

What is the ratio $OR : RB$?

A 1 : 2 **B** 1 : 1 **C** 3 : 2

D 1 : 3 **E** 2 : 3 **F** 3 : 5

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15. The period T of the natural oscillations of a small liquid droplet depends only on the density ρ of the liquid (in kg m^{-3}), the radius r of the droplet (in m) and the surface tension σ of the liquid (in N m^{-1}, where 1 N = 1 kg m s^{-2}), according to a formula

$$T = k\rho^a r^b \sigma^c,$$

where k is a number with no units and a, b and c are constants.

A droplet of radius 2 mm oscillates with period 8 ms. What is the period of a droplet of the same liquid with radius 0.5 mm?

A 16 ms **B** 0.25 ms **C** 1 ms

D 0.5 ms **E** 2 ms **F** 4 ms

16. A parking meter accepts only 10p, 20p and 50p coins, which are inserted one at a time. A driver wants to insert coins with a total value of exactly £1. Two ways of doing this count as different if the sequences of coins are different; for example, inserting 50p, 20p, 20p, 10p in that order is different from inserting 20p, 50p, 20p, 10p.

How many different ways are there?

A 10 **B** 64 **C** 127

D 89 **E** 144 **F** 128

17. A game starts with a score of 1. On each turn a fair coin is tossed: if it shows heads the score is doubled, and if it shows tails 1 is added to the score. The coin is tossed six times in all.

What is the probability that the final score is a multiple of 3?

A $\frac{5}{16}$ **B** $\frac{11}{64}$ **C** $\frac{1}{3}$

D $\frac{21}{64}$ **E** $\frac{3}{8}$ **F** $\frac{11}{32}$

18. A rectangular rug measuring 4 m by 3 m lies flat on a floor. Keeping it flat on the floor, it is turned through 90° about one of its corners, which stays fixed.

What is the total area of floor, in m^2, that is covered by the rug at some time during this movement, including its starting and finishing positions?

A $24 + \frac{25\pi}{4}$ **B** $12 + \frac{25\pi}{4}$ **C** $\frac{25\pi}{2}$

D $6 + \frac{25\pi}{4}$ **E** $12 + 4\pi$ **F** $\frac{25\pi}{4}$

19. Ana, Ben and Cat choose one of themselves by the following method. In each round, all three toss their own coins at the same time. If one person's coin shows a different face from the other two coins, that person is chosen; if all three coins show the same face, they play another round.

Ana's coin shows heads with probability $\frac{1}{3}$, Ben's coin shows heads with probability $\frac{1}{4}$, and Cat's coin is fair. All tosses are independent. What is the probability that Cat is chosen?

A $\frac{7}{17}$ **B** $\frac{1}{3}$ **C** $\frac{7}{24}$

D $\frac{7}{12}$ **E** $\frac{5}{17}$ **F** $\frac{5}{12}$

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20. A sequence is defined by $u_{n+1} = u_n^2 - 2$ for $n \geq 1$, where the first term u_1 is a real number. For certain values of u_1 the sequence alternates between two different values, so that $u_3 = u_1$ but $u_2 \neq u_1$.

What is the largest such value of u_1?

A $\sqrt{3}$ **B** $\sqrt{2}$ **C** -1

D $\dfrac{\sqrt{5}-1}{2}$ **E** 2 **F** $\dfrac{1+\sqrt{5}}{2}$

21. Ten seats in an assembly are shared among three parties by the following method. The seats are awarded one at a time. Before each award, each party's *quotient* $\dfrac{V}{s+1}$ is calculated, where V is the party's number of votes and s is the number of seats it has already been awarded; the next seat goes to the party with the largest quotient, and if two quotients are equal it goes to the party with more votes.

Party A receives 4200 votes, party B receives 3000 votes and party C receives 1800 votes, and C wins 2 seats. If the votes for A and B had been unchanged, what is the smallest number of extra votes that C would have needed in order to win 3 seats?

A 451 **B** 1286 **C** 1201

D 720 **E** 721 **F** 901

22. A storeroom is a cube with edges of length 6 m. A taut straight wire runs from one corner of the floor to the corner of the ceiling furthest from it. A thin straight pipe runs along one of the vertical edges of the room that the wire does not touch.

What is the shortest distance between the wire and the pipe, in metres?

A $2\sqrt{6}$ **B** 3 **C** $3\sqrt{2}$

D $3\sqrt{3}$ **E** $2\sqrt{3}$ **F** $\sqrt{6}$

23. In this question, the quartiles of an ordered list of numbers are the medians of its lower and upper halves; when the list has an odd number of entries, the middle entry is left out of both halves. An entry in the list is called an *outlier* if it is more than 1.5 times the interquartile range above the upper quartile, or more than 1.5 times the interquartile range below the lower quartile.

Seven runners have times, in seconds, of

$$30, \quad 34, \quad 35, \quad 38, \quad 41, \quad 42, \quad 50.$$

An eighth runner's time, x seconds, is added to the list, where x is a positive integer. For how many values of x is x *not* an outlier in the new list of eight times?

A 45 **B** 46 **C** 47

D 42 **E** 37 **F** 33

24. Four fair six-sided dice, each numbered 1 to 6, are rolled. What is the probability that the scores on some two of the dice add up to 7?

A $\dfrac{1}{2}$ **B** $\dfrac{125}{216}$ **C** $\dfrac{2}{3}$

D $\dfrac{77}{216}$ **E** $\dfrac{91}{216}$ **F** $\dfrac{139}{216}$

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25. A square has two of its vertices on the curve $y = x^2$ and its other two vertices on the line $y = x - 4$. What is the sum of the areas of all such squares?

A 100 **B** 36 **C** 68

D 34 **E** 18 **F** 50

26. Five children, all of different heights, stand in a row for a photograph, in a random order with all orders equally likely. The photographer calls the row *jagged* if each of the three children who are not at the ends of the row is either taller than both of their neighbours or shorter than both of their neighbours.

What is the probability that the row is jagged?

A $\frac{1}{3}$ **B** $\frac{1}{5}$ **C** $\frac{1}{4}$

D $\frac{3}{10}$ **E** $\frac{2}{15}$ **F** $\frac{4}{15}$

27. A machine takes a positive integer n, writes it as a product of powers of distinct primes, and replaces each prime p by $p + 1$, keeping the same powers, to produce a number $f(n)$. For example, $12 = 2^2 \times 3$, so $f(12) = 3^2 \times 4 = 36$, and $f(7) = 8$. The machine also gives $f(1) = 1$.

For how many positive integers $n \leq 1000$ is $f(n)$ a multiple of n?

A 12 **B** 10 **C** 8

D 5 **E** 9 **F** 6

End of paper

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